

Agentic AI Harness Roadmap

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Free LLM & AI API Platforms

No credit card required, permanent free tier

- **Google AI Studio** — Gemini Flash models, up to 1M tokens of context at lower request volume. Uses your prompts for training unless you're in the EEA, UK, or Switzerland.
- **Groq** — known for very high inference speed on Llama/Qwen models, smaller model catalogue.
- **Cerebras** — also high throughput, similar positioning to Groq.
- **OpenRouter** — one key across many providers' free models; free pool capped around 50 requests/day (failed requests still count against it); limits aren't a fixed guarantee.
- **NVIDIA NIM** (build.nvidia.com) — OpenAI-compatible key, 120+ open-weight models; NVIDIA restricts it to trial/dev-and-test use and logs usage for its own product improvement.
- **Hugging Face Inference** — free tier for hosted inference on many open models.
- **Cloudflare Workers AI** — free allotment for inference.
- **Mistral** — permanent free tier alongside its paid offering.
- **Eden AI** — free LLM models via the same platform as its paid models, including Google-hosted models with up to 262K context (EU region) and select Cloudflare-hosted models.

No key needed at all (anonymous, rate-limited)

- **OVHcloud AI Endpoints** — permanent free anonymous tier, 2 requests/min per IP per model, no signup; an API key unlocks higher limits (400 RPM per project), billed pay-as-you-go with \$200 in trial credits for new accounts.
- **Kilo Code** — free router pool; docs warn it may route requests to providers that log prompts and outputs.
- **LLM7.io** — anonymous access confirmed working; catalogue changes often.

Notes on reliability

- **Together AI** dropped its free trial — new accounts now require a minimum \$5 prepaid balance.
- **SiliconFlow** now requires real-name ID verification for free models.
- Maintained community reference: GitHub repo [mnfst/awesome-free-](https://github.com/mnfst/awesome-free-ai)

llm-apis (live-tests providers regularly).

Free-Tier Business Tool APIs (Email, CRM, Automation)

- **Brevo** — 300 emails/day (9,000/month), full API access, no credit card required; bundles CRM, SMS, and marketing automation.
- **Amazon SES** — 3,000 emails/month free (more technical setup required).
- **Mailtrap** — up to 4,000 emails/month, capped at 150/day.
- **SendGrid** — replaced its permanent free tier with a 60-day trial in 2025; paid plans start at \$19.95/month after that (verify current terms before relying on it).
- **Resend** — 3,000 emails/month free with full API.
- **Mailjet** — 6,000 emails/month combined (transactional + marketing).
- **HubSpot** — unlimited free CRM contacts, plus 2,000 email sends/month.
- **SMTP2GO** — 1,000 free emails/month.
- **SendPulse** — 15,000 emails/month free.
- **Loops** — 1,000 contacts with event-driven automations, useful for SaaS onboarding flows.

Free-tier terms (rate limits, card requirements, verification steps) change frequently — confirm current terms on each provider's signup page before building on one.

Roadmap: Agentic AI Harness Using the Above Tools

Builds on existing work: the Anthropic API + Microsoft Graph meeting-scheduling agent, and the 135-topic AI engineering roadmap tracker.

Phase 1 — Core Harness Skeleton (Week 1-2)

Goal: one Python process that can call any LLM provider through a single interface.

- Build a provider-agnostic client wrapper — a class with `.chat(messages, model, tools)` that routes to Anthropic, Groq, Google AI Studio, or OpenRouter based on a config flag. Groq, OpenRouter, and NVIDIA NIM are OpenAI-compatible, so one adapter covers them with just a different `base_url` and key.
- Store keys in `.env`, load with `python-dotenv`. Never hardcode credentials.
- Implement a basic agent loop: system prompt → user input → model response → check for `tool_use` → execute → feed result back → repeat until final answer.
- Test the same loop against 2-3 providers (e.g., Anthropic for reasoning, Groq for speed) to confirm the abstraction is actually provider-agnostic.

Phase 2 — Multi-Provider Routing & Fallback (Week 2-3)

Goal: resilience — if one free tier is rate-limited, the harness fails over automatically.

- Add a router that tries providers in priority order (e.g., Groq → Google AI Studio → OpenRouter free pool → NVIDIA NIM), catching 429/rate-limit errors to fall through to the next.
- Log which provider actually served each request, to identify which free tiers are reliable enough to depend on.
- Add simple token/cost tracking even though it's free — this primes the harness for when paid tiers are eventually mixed in.

Phase 3 — Tool / Function-Calling Layer (Week 3-4)

Goal: give the agent hands, not just a mouth.

- Define a small tool registry (name, description, JSON schema, Python function) — the same pattern Anthropic/OpenAI tool-calling expects, transferable across providers with minor schema tweaks.
- Start with low-risk tools: web search, a calculator, a file reader.
- Add Microsoft Graph tools next, building on the existing integration — calendar read/write, email search.

Phase 4 — Business-Tool Integrations (Week 4-6)

Goal: turn the agent into something that can actually do office work.

- **Email action tool:** wire in Brevo or Resend's free tier so the agent can draft and send transactional emails (start with draft-only + human approval).
- **CRM tool:** HubSpot's free tier (unlimited contacts) as a place for the agent to look up or log contact/deal info.
- **Automation trigger tool:** Loops or SendPulse for event-driven sequences (e.g., agent detects a task completion and triggers a notification workflow).
- Each becomes one more entry in the Phase 3 tool registry — same interface, different backend.

Phase 5 — Memory & State (Week 6-7)

Goal: the agent remembers across turns and sessions, not just within one call.

- Start simple: a JSON or SQLite store keyed by conversation/session ID (ties into existing SQL/DB fundamentals study).
- Add a lightweight retrieval step: before each turn, pull relevant past context and inject it into the system prompt.
- Consider formalizing the tracker-and-status pattern already used for the roadmap Excel sheet, but as agent-readable state instead of

a spreadsheet.

Phase 6 — Orchestration & Multi-Agent (Week 7-9)

Goal: move from one agent to a small team of specialized agents.

- Split responsibilities: a planner agent (decides what needs doing), an executor agent (calls tools), and optionally a reviewer agent (checks executor output before it goes out — e.g., before an email actually sends).
- Route planner/reviewer calls to a stronger model (Anthropic, or Google AI Studio's 1M-context Gemini) and high-volume executor calls to a fast/free model (Groq, Cerebras) — mixing direct and routed access.
- Add a simple task queue so the agent can work asynchronously instead of blocking on each tool call.

Phase 7 — Observability & Guardrails (Week 9-10)

Goal: make it safe and debuggable enough to actually rely on.

- Log every tool call, its inputs, and outputs to a file or SQLite table — this becomes the audit trail, especially important once email-send or CRM-write tools are live.
- Add a human-in-the-loop confirmation step for any action tool that sends something externally.
- Add basic rate-limit-aware backoff so retries don't burn through free-tier quotas.

Phase 8 — Packaging (Week 10+)

- Wrap the harness as a CLI or a small FastAPI service so it can be called like a product, not just run as a script.
- Push it to GitHub as a portfolio piece for the AI engineering track — ties directly into the existing profile README.

Suggested first milestone: get Phases 1-3 working end-to-end with just Anthropic + Groq + one Microsoft Graph tool, since working pieces of both already exist.